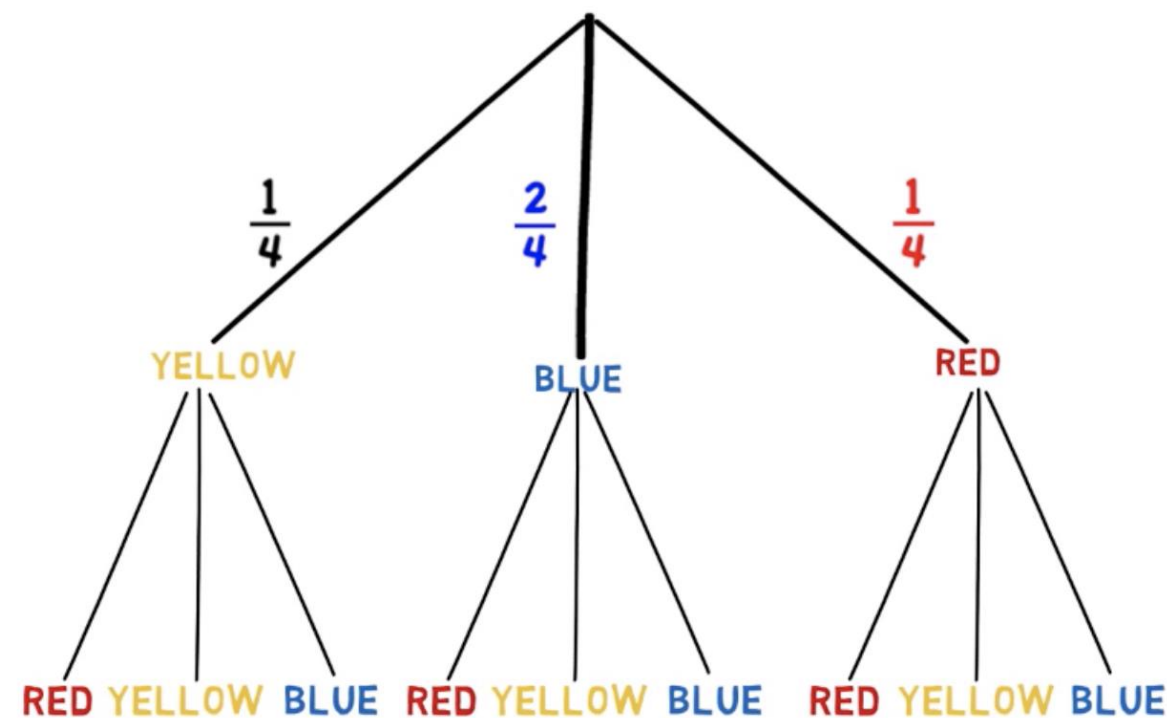


CS 237: PROBABILITY IN COMPUTING

INSTRUCTORS: NATHAN KLEIN AND TIAGO JANUARIO

BOSTON
UNIVERSITY

STEP 1: CREATE PROBABILITY TREE



From <https://www.youtube.com/watch?v=UxiCV8RroKU>

LECTURE 4

Last time

- Why do you need the axioms?
- Tree diagrams
- Analysis of the Monty Hall problem
- Dice game

Today

- More axioms and probability rules
- Continuous Probability Spaces
- Anomalies with Continuous Probability

GET READY FOR VOTING!

- ▶ Students from A1 section:
 1. Enter this URL: <https://app.tophat.com/e/779973>
 2. Login as needed using your Top Hat account info.
 3. If asked about enrolling, click *Enroll*.

- ▶ Students from A2 section:
 1. Enter this URL: <https://app.tophat.com/e/544135>
 2. Login as needed using your Top Hat account info.
 3. If asked about enrolling, click *Enroll*.

ASK THE AUDIENCE

Toss a coin twice. Which outcomes did you get?

A. HH

B. TT

C. Two different outcomes.

ASK THE AUDIENCE

A family has 2 children. Assume that each child is equally likely to be a boy or a girl. Which of the following is more likely?

A. They have 2 boys.

B. They have 2 girls.

☒ C. They have 2 kids of different gender.

D. All the three possibilities above are equally likely.

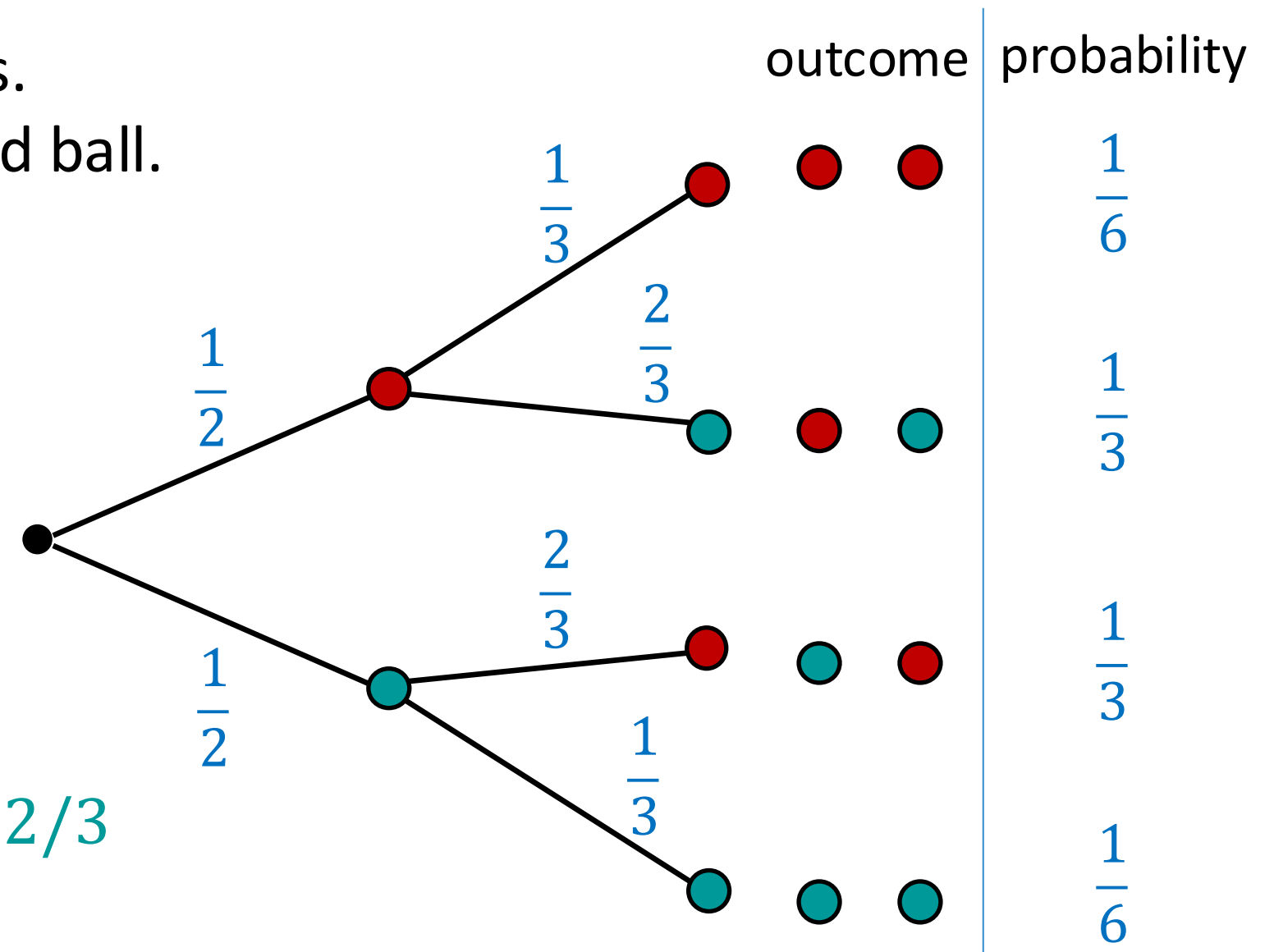
$$\frac{1}{4} \quad \{gg, bb, bg, gb\}$$
$$\frac{1}{4}$$

$$E = \{bg, gb\}$$
$$Pr[E] = \frac{1}{2}$$

ASK THE AUDIENCE

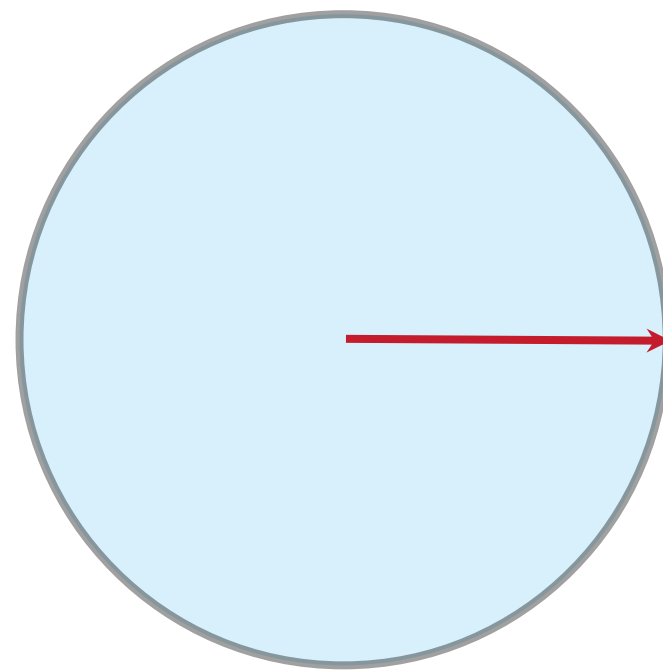
You have a bag containing **red** and **green** balls.
 You draw one ball from the bag, then a second ball.
 The tree diagram shows the experiment.
 Consider the following three statements.

1. Drawing was without replacement
2. The bag originally contained 2 red and 2 green balls
3. $\Pr(\text{the 2 balls are of different colors}) = 2/3$

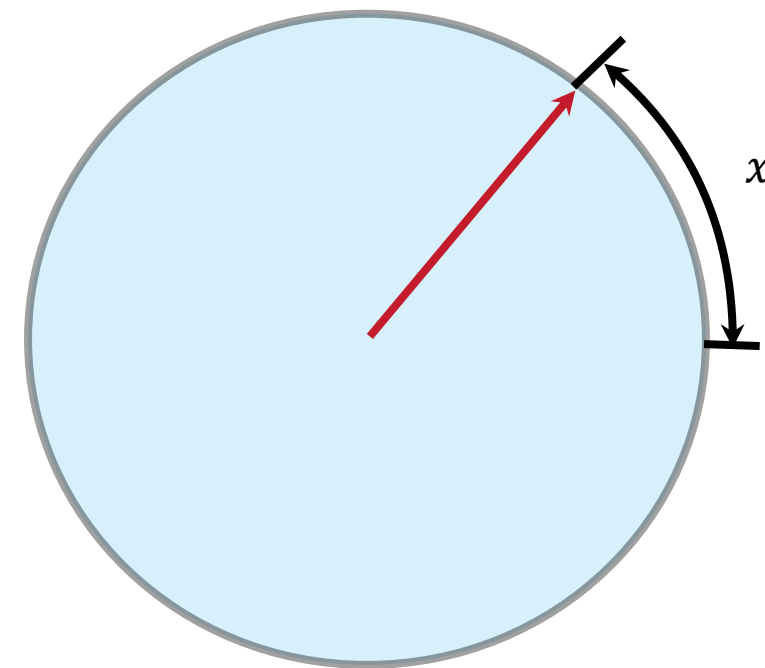


(A) 1 is true B) 2 is true C) 3 is true (D) all are true

SPINNERS



spin dial

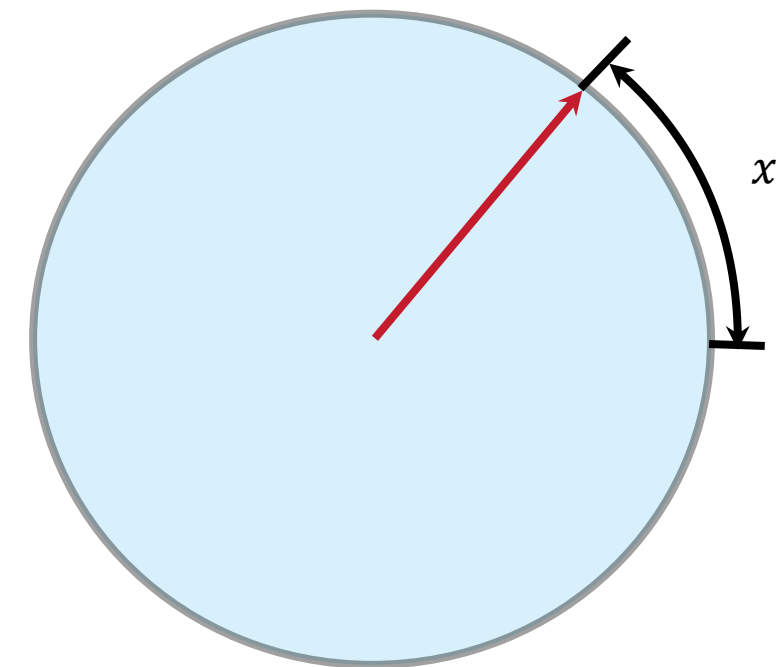


Perimeter = 1

$$\text{radius} = \frac{1}{2\pi}$$

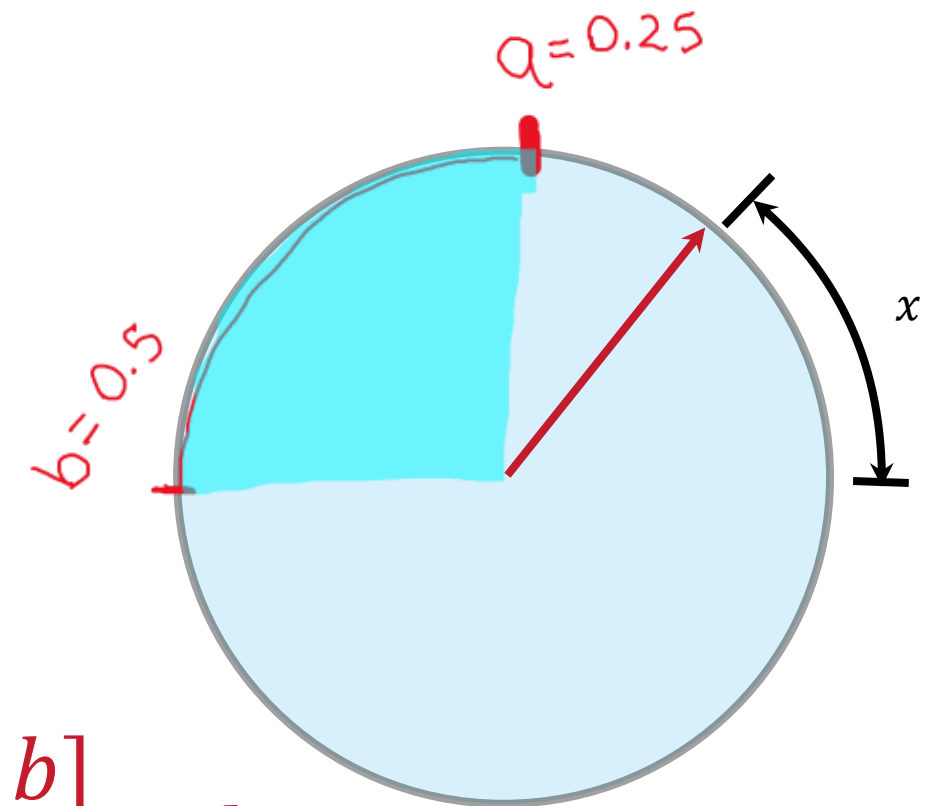
SPINNERS

- ▶ **Experiment:** spin the dial
- ▶ **Outcome:** real number $x \in [0,1]$
- ▶ **Sample space:** $\Omega = [0,1)$

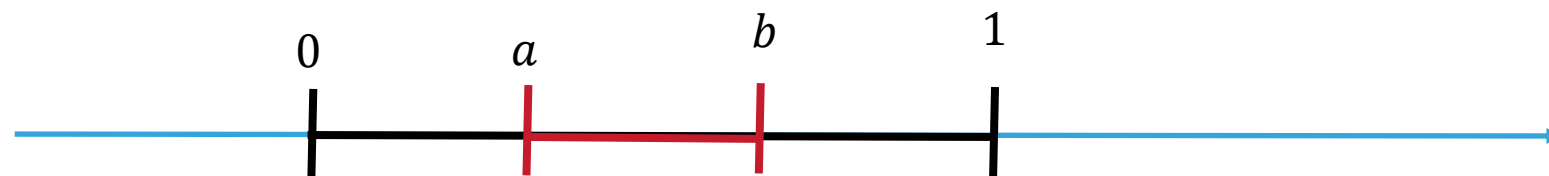


SPINNERS P_v ["spinner stops in $[0.25, 0.5]$] = $0.5 - 0.25 = 0.25$

- ▶ **Experiment:** spin the dial
- ▶ **Outcome:** real number $x \in [0,1)$
- ▶ **Sample space:** $\Omega = [0,1)$
- ▶ **Probability function:** for all $[a, b] \subseteq [0,1]$



$$Pr(x \in [a, b]) = \frac{\text{length of } [a, b]}{\text{length of } [0, 1]} = b - a$$



ASK THE AUDIENCE

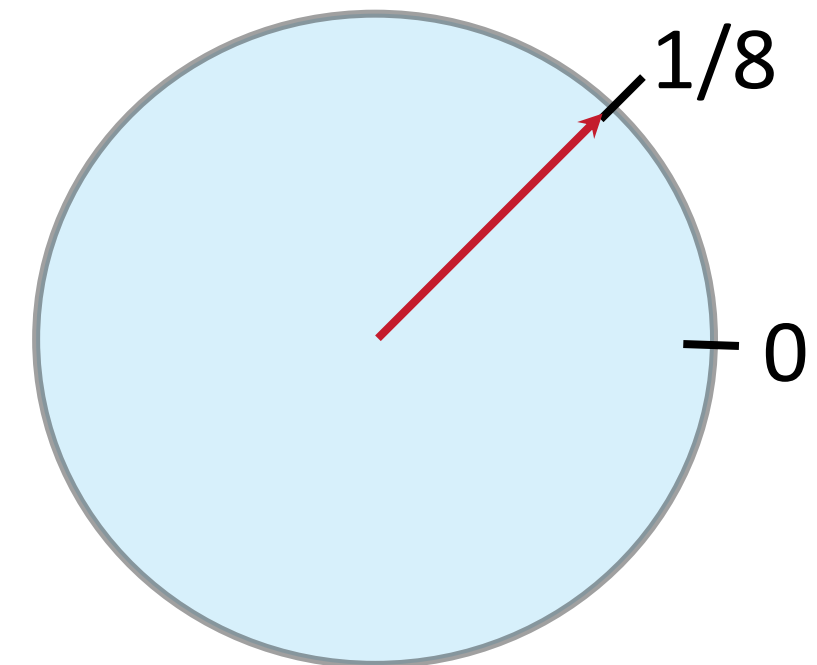
- What is the probability that the dial lands on $1/8$?

(A) $1/8$

(B) 0

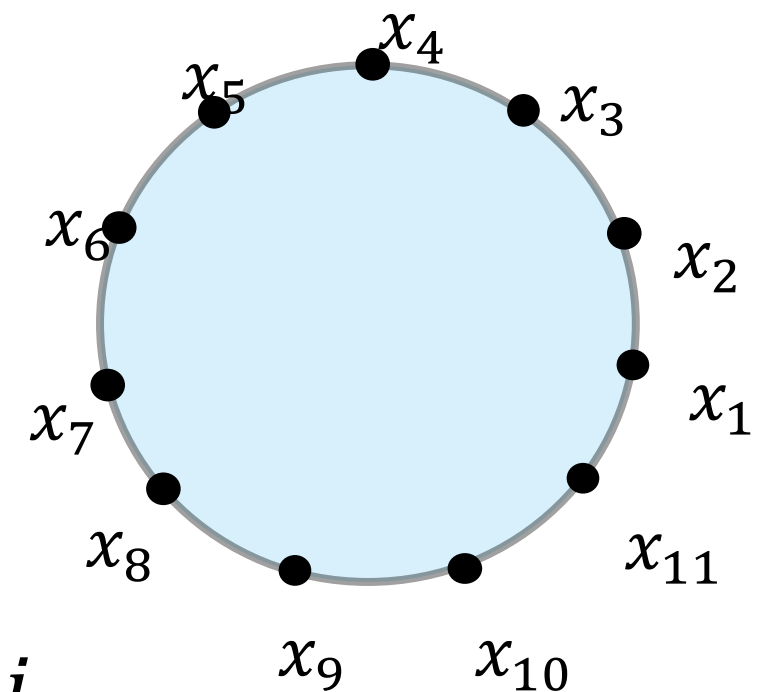
(C) 0.01

(D) none of the above



SPINNERS

- ▶ What is the probability that the dial lands on $1/8$?
- ▶ Suppose we suspect that it is, say, 0.1
- ▶ Pick 11 distinct points on the circle



by symmetry: $Pr(\text{dial lands on } x_i) = 0.1$ for all i

by additivity: $Pr(\text{dial lands on one of } x_1, \dots, x_{11}) = 11 \cdot 0.1 = 1.1$

contradiction!

SPINNERS

- ▶ Let p be the probability of an outcome, e.g., $1/8$
- ▶ We can show that $p = 0$ by contradiction
- ▶ If $p \neq 0$: pick distinct points $x_1, \dots, x_k$, where $k > 1/p$

by symmetry: $Pr(\text{dial lands on } x_i) = p \ \forall i \in \{1, 2, \dots, k\}$

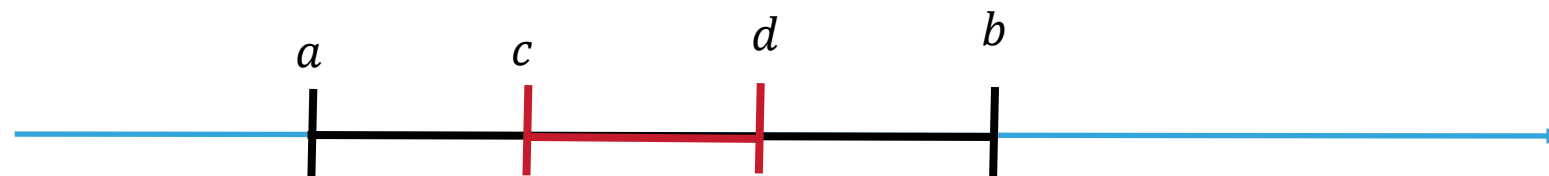
by additivity: $Pr(\text{dial lands on one of } x_1, \dots, x_k) = k \cdot p > 1$

contradiction!

UNIFORM SAMPLING FROM AN INTERVAL

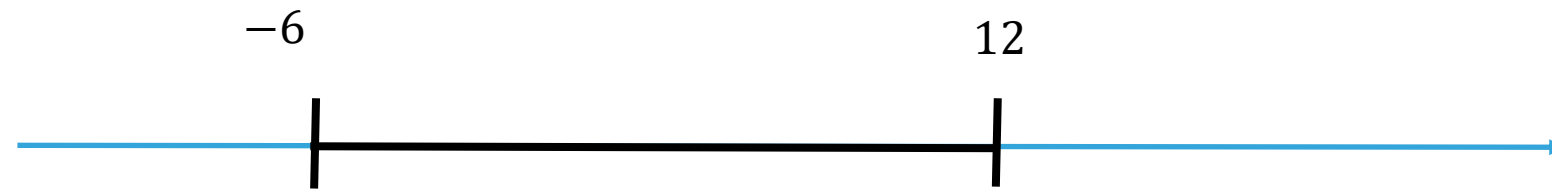
- ▶ **Experiment:** sample a point uniformly from $[a, b]$
- ▶ **Outcome:** real number $x \in [a, b]$
- ▶ **Sample space:** $\Omega = [a, b]$
- ▶ **Probability function:** for all $[c, d] \subseteq [a, b]$, we have

$$\Pr(x \in [c, d]) = \frac{\text{length of } [c, d]}{\text{length of } [a, b]} = \frac{d - c}{b - a}$$



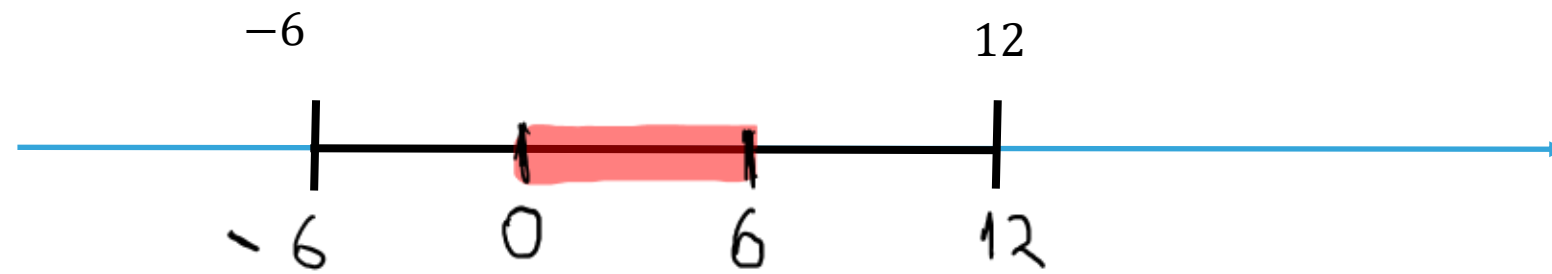
UNIFORM SAMPLING FROM AN INTERVAL

$$x \in \text{Uniform}([-6, 12])$$



UNIFORM SAMPLING FROM AN INTERVAL

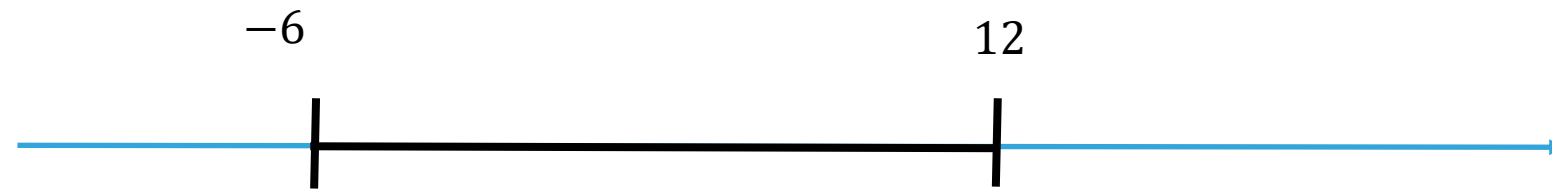
$$x \in \text{Uniform}([-6, 12])$$



$$\text{Pr}(x \in [0, 6]) = \frac{6 - 0}{12 - (-6)} = \frac{6}{18} = \frac{1}{3}$$

UNIFORM SAMPLING FROM AN INTERVAL

$$x \in \text{Uniform}([-6, 12])$$



$$\text{Pr}(x \in [0, 5)) = \text{Pr}(x \in [0, 5]) = \frac{5 - 0}{12 - (-6)} = \frac{5}{18}$$

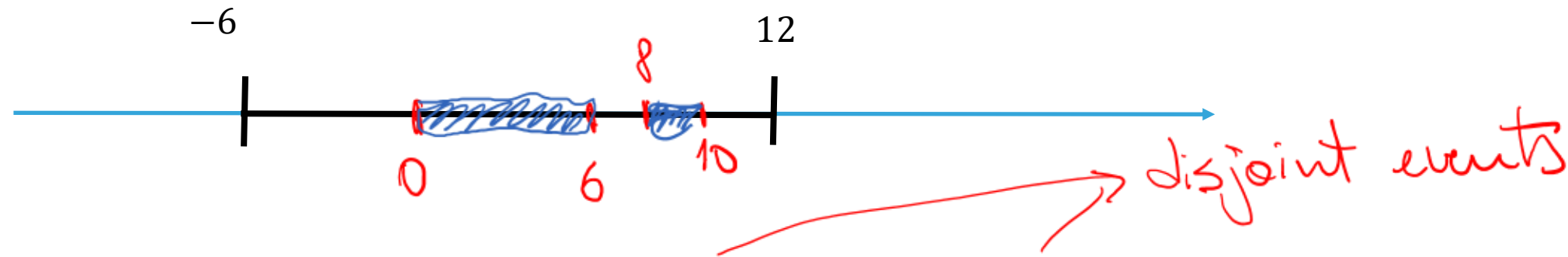
$$\Downarrow$$

$$\text{Pr}(x \in [0, 5)) + \text{Pr}(x = \{5\}), \text{ disjoint events}$$

$$\frac{5}{18} + 0 = \frac{5}{18}$$

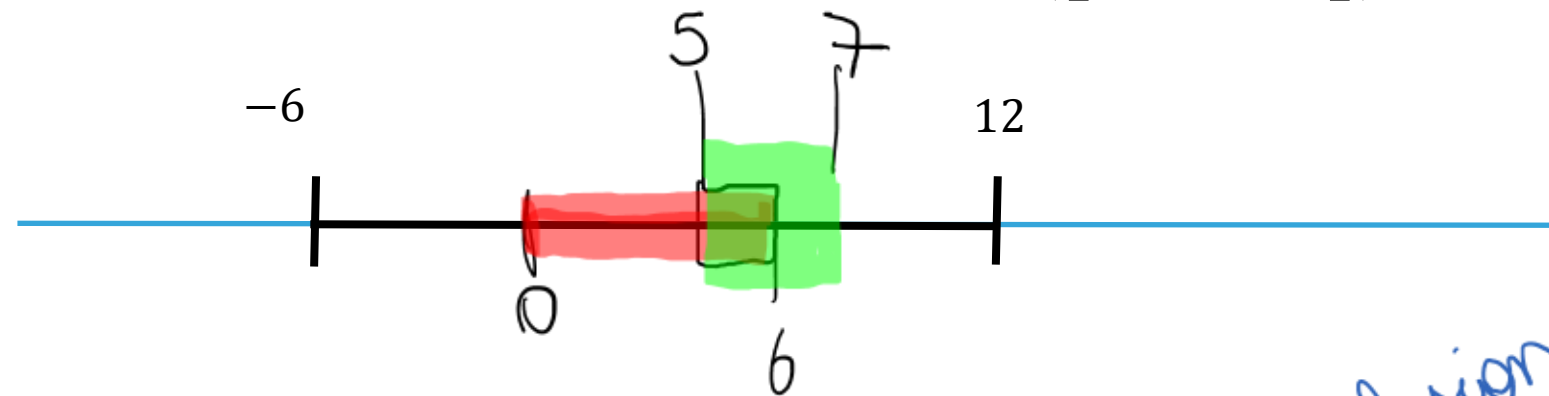
UNIFORM SAMPLING FROM AN INTERVAL

$$x \in \text{Uniform}([-6, 12])$$



$$\begin{aligned}
 \text{▶ } Pr(x \in [0, 6] \cup [8, 10]) &= Pr(x \in [0, 6]) + Pr(x \in [8, 10]) \\
 &= \frac{6}{18} + \frac{2}{18} = \frac{8}{18} = \frac{4}{9}
 \end{aligned}$$

UNIFORM SAMPLING FROM AN INTERVAL

 $x \in \text{Uniform}([-6, 12])$ 

► $\Pr(x \in [0, 6] \cup [5, 7]) = \frac{7}{18}$

use the inclusion-exclusion principle

$$\Pr(x \in [0, 6]) + \Pr(x \in [5, 7]) - \Pr(x \in [5, 6])$$

$$= \frac{6}{18} + \frac{2}{18} - \frac{1}{18} = \frac{7}{18}$$

UNIFORM SAMPLING FROM AN INTERVAL

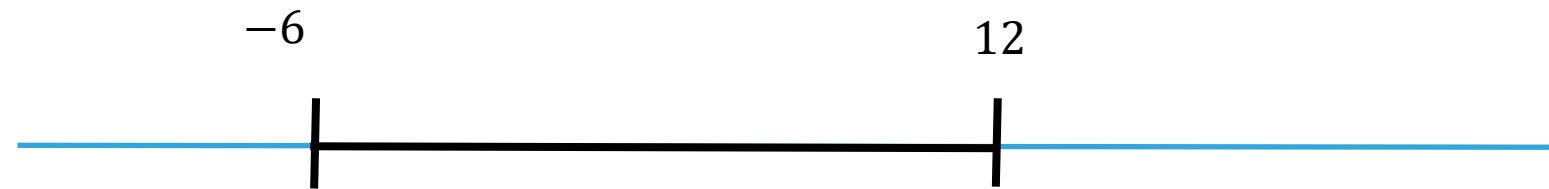
$$x \in \text{Uniform}([-6, 12])$$



$$\begin{aligned}
 \text{► } Pr(x \in [-10, 10]) &= \frac{10 - (-6)}{12 - (-6)} = \frac{16}{18} \\
 &= Pr(x \in [-10, -6)) \cup Pr(x \in [-6, 10]) \\
 &= 0 + \frac{16}{18} = \frac{16}{18}
 \end{aligned}$$

UNIFORM SAMPLING FROM AN INTERVAL

$$x \in \text{Uniform}([-6, 12])$$



- ▶ $Pr(x \in [0, 6]) =$
- ▶ $Pr(x \in [0, 5)) =$
- ▶ $Pr(x \in [0, 6] \cup [8, 10]) =$
- ▶ $Pr(x \in [0, 6] \cup [5, 7]) =$
- ▶ $Pr(x \in [-10, 10]) =$

UNIFORM SAMPLING FROM AN INTERVAL

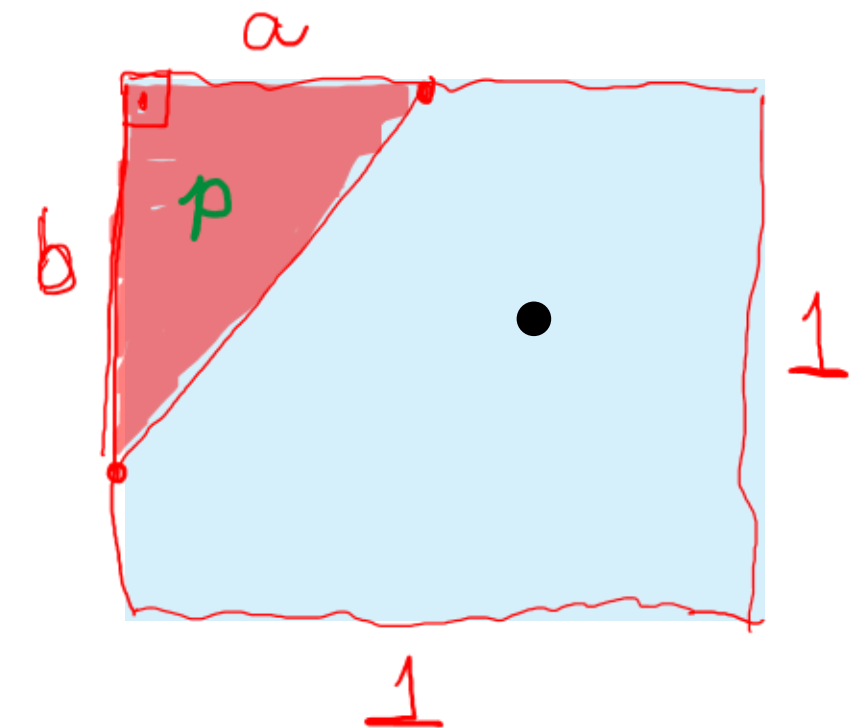
```
from numpy.random import uniform
```

```
# uniform sampling from [0, 1]  
def uniform_unit_interval():  
    x = uniform(0, 1)  
    return x
```

```
# uniform sampling from [a, b]  
def uniform_interval(a, b):  
    x = uniform(a, b)  
    return x
```

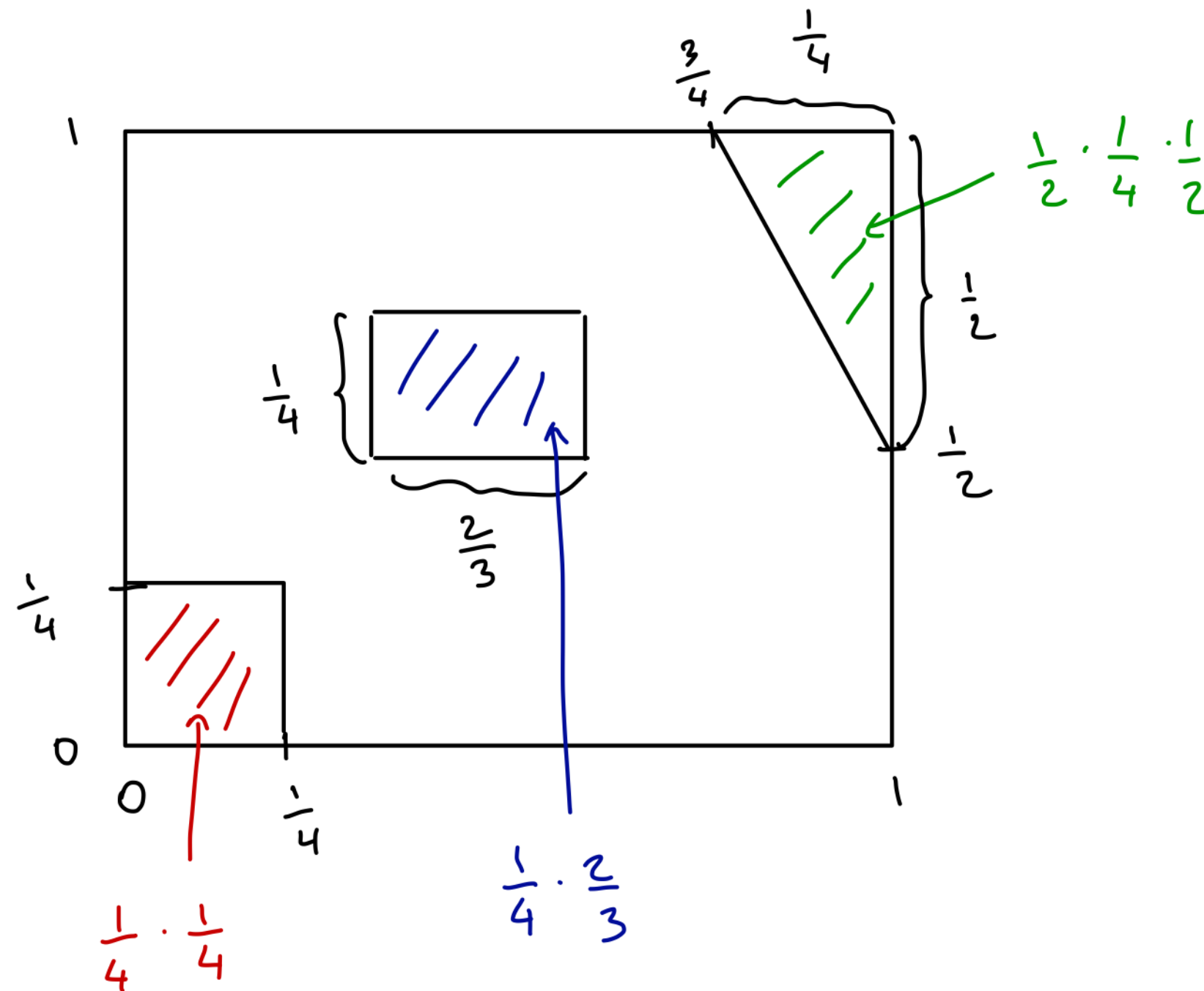
DARTS WITH A SQUARE BOARD

- ▶ **Experiment:** throw a dart at a square board
- ▶ **Outcome:** hitting point = Probability is zero
- ▶ **Sample space:** the square
- ▶ **Probability function:** $Pr(p \in P) = \frac{\text{area of } P}{\text{area of square}}$



$$Pr(\text{"of hitting } p\text{"}) = \frac{\text{area of } p}{\text{area square}} = \frac{a \cdot b}{2}$$

UNIFORM SAMPLING FROM THE UNIT SQUARE



UNIFORM SAMPLING FROM RECTANGLES

```
# uniform sampling from unit square [0, 1]x[0, 1]
def uniform_unit_square():
    x = uniform(0, 1)
    y = uniform(0, 1)
    return (x, y)

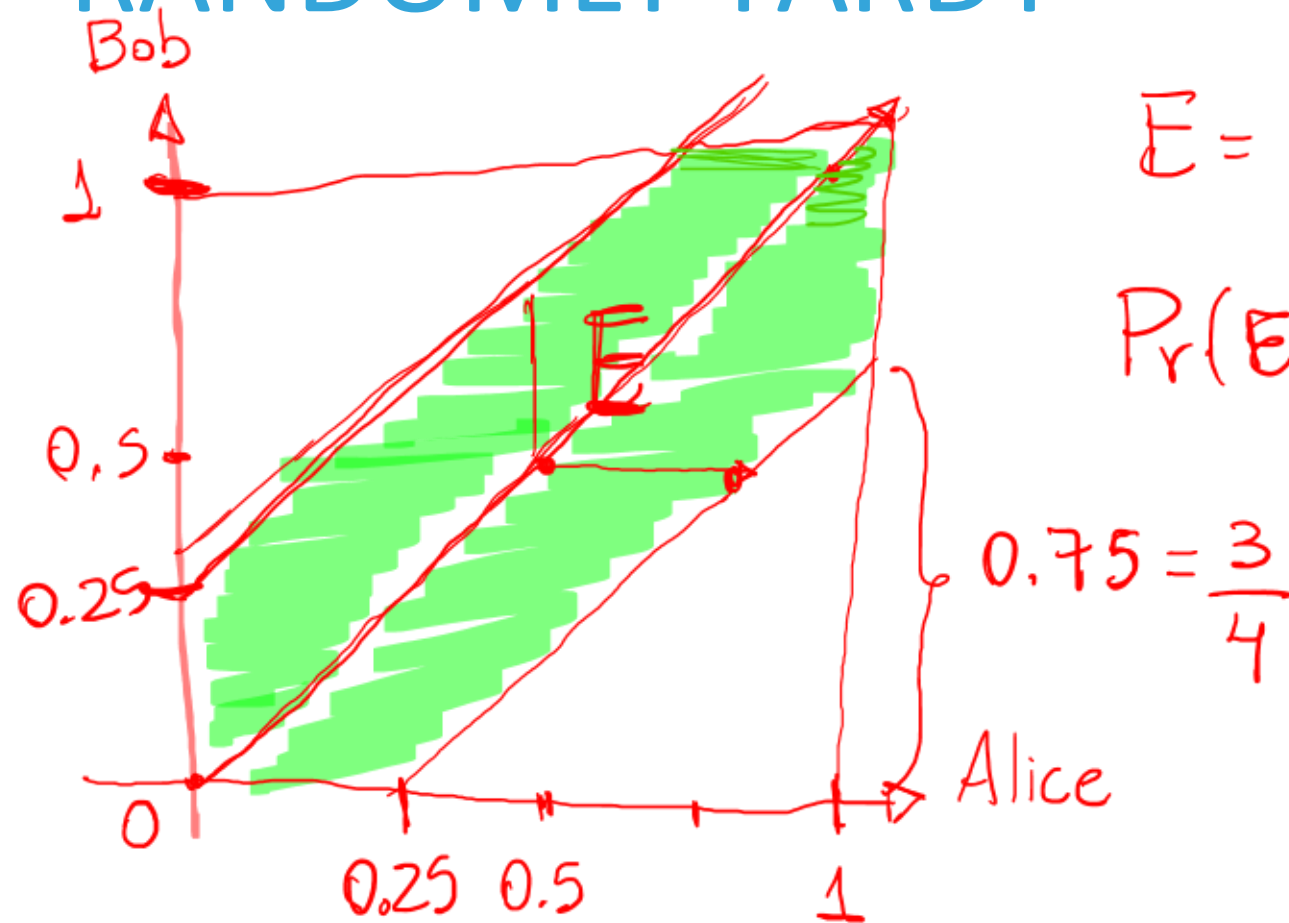
# uniform sampling from the rectangle [a, b]x[c, d]
def uniform_rectangle(a, b, c, d):
    x = uniform(a, b)
    y = uniform(c, d)
    return (x, y)
```

RANDOMLY TARDY

- ▶ Alice and Bob decide to meet for a Pokemon Go raid at noon
- ▶ Each of them arrives at a time chosen uniformly at random between noon and 1pm
- ▶ Once one of them arrives, they will wait for the other for 15 minutes and leave if they do not show up
- ▶ What is the probability that Alice and Bob do the raid together?

RANDOMLY TARDY

(Math on the board)

 $\bar{E}$ = event that Bob and Alice meet

$$\Pr(E) = 1 - \Pr(\bar{E}) = 1 - \left(\frac{3}{4}\right)^2$$

$$= 1 - \frac{9}{16} = \frac{16}{16} - \frac{9}{16} = \frac{7}{16}$$

ASK THE AUDIENCE

We toss a coin 5 times. The outcome is a string of length 5 of H's and T's, where H means ``heads'' and T means ``tails''.

How many outcomes are there where the first toss is H **or** we have exactly 2 T's?

- A. 6
- B. 10
- C. 16
- D. 20
- E. None of the above